# Dustin Biser

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## Interests and Focus

Computer Graphics, Rigid Body Simulation, Computational Geometry, and Numerical Methods.

## Education

University of Maryland, College Park
Bachelor of Science in Physics, 2012
Bachelor of Science in Mathematics, 2012
Minor in Computer Science

## Recent Projects

3D Graphics Engine Link
*July 2013 – Present*
Engine for real-time 3D graphics. Developed using C++, OpenGL, and GLSL.

Physics Engine Link
*December 2011 – May 2012*
My senior project in college, which involved creating a 2D physics engine for convex rigid bodies.

## Employment

Computer Vision Software Engineer
*June 2012 – March 2013*
*SAIC (Arlington, VA)*

- C++, OpenGL, Qt development on massively simulated 3D graphical application used for military planning and training.
- Java, OpenGL development on multithreaded application used for stabilizing and stitching together video frames from unmanned aerial vehicles in real time.

Software Engineer II
*May 2013 – September 2013*
*Appian (Reston, VA)*

- Java development in fast pace agile environment aimed at creating Business Process Management (BPM) tools for businesses in order to boost their overall efficiency.
- Designed, implemented, and tested new features for cutting edge web and mobile BPM applications.

Web Programmer
*May 2010 – Aug 2011*
*Optoro (Lanham, MD)*

- Ruby on Rails development for an in-house product management system used for receiving, selling, and shipping online merchandise.
- Lots of work involving Databases, Git, Agile Development, and Test Driven Development.

## Research Experience

Undergraduate Research Intern
*May 2009 – August 2009*
*National Institute of Standards and Technology (Gaithersburg, MD)*
Developed an interactive 3D graphical solution for interpolating and displaying the amounts of radiation delivered throughout containers that had been sent through industrial irradiation facilities.

Undergraduate Research Intern
*May 2008 – August 2008*
*National Institute of Standards and Technology (Gaithersburg, MD)*
Created and validated a software controller for programming electro-optical devises used in high-fidelity remote sensing instrument calibrations.

## Main Skills

- Strong 3D graphics theory knowledge
- Strong Math and Physics knowledge
- Excellent written and spoken communication skills
- Experience with multithreaded programming and architectures
- Modeling: 3D Mesh Creation and UV Mapping
- Fluent in C/C++, GLSL, and Java.
- Libraries: OpenGL, Boost, STL, GLM, Qt
- Graphics Tools: Blender, Gimp, Photoshop
- Development: Visual Studio, Eclipse, git, subversion, premake4, cmake

## Open Software Contributions

OpenGL Mathematics (GLM) Link
*October 2013 – Present*
So far I have contributed bug fixes and optimizations for quaternion routines. I plan to continue contributing to this project in the future as it is very useful and has gained much popularity within the graphics community.